GAJENDRA DHANOLIYA

+91-9109485566 | Gmail | Portfolio | LinkedIn | GitHub | Codeforces | CodeChef

Education

Indian Institute of Technology Delhi 2022 – 2026
B.Tech in Electrical Engineering

Jawahar Navodaya Vidyalaya Sitapura, Bundi, Rajasthan 2020 – 2021
CBSE Class XII

Jawahar Navodaya Vidyalaya Joura, Morena, Madhya Pradesh 2018 – 2019
CBSE Class X

Scholastic Achievements

- **JEE Advanced:** Secured All India Rank in the top **0.5%** out of **1M+** candidates nationwide 2022
- **Competitive Programming:** Codeforces **Specialist** (**1500+**, G.Dhanoliya) and CodeChef **4 Star** (**1800+**, gajenx7) and solved **1000+** problems across competitive programming platforms Present

Experience

Software Engineer Intern | ZeTheta Algorithms Pvt. Ltd. Jun 2025 – Aug 2025

- Built an automated **eKYC** pipeline that handles document capture and **Aadhaar** based identity verification, guiding applicants through each step and cutting manual onboarding time by **60%**
- Developed the **Video KYC** check, where **SSD ResNet-10** finds the face in each frame, **dlib** turns it into a **128-d embedding** compared to the ID photo by **Euclidean distance**, and a **passive anti spoof CNN** rejects printed photos and screen replays
- Tuned the match threshold on labelled ID and selfie pairs, combined the results from several frames and ignored blurry or poorly lit ones, reaching **93% face verification accuracy**

Projects

RouteOS | *Python, FastAPI, OR-Tools, Redis, React, AWS* | GitHub

- Built a full stack multi vehicle route optimization platform solving the **capacitated VRP** in **Google OR Tools**, modelling **time windows**, service durations and per vehicle distance caps as constraint dimensions with priority scaled drop penalties
- Cut total fleet travel distance by **20–35%** against a **greedy nearest neighbour** baseline solved on identical order sets (**375 km vs. 576 km** on 100 orders across 12 vehicles) with **100% order assignment** on **50–250** order benchmarks
- Streamed live vehicle simulation over **WebSockets** with traffic injection and **rolling horizon re-optimization** that re-solves undelivered stops from the vehicle's live position, backed by **PostGIS** proximity queries and **Redis** cached analytics
- Deployed on **AWS EC2** (**VPC** public subnet, **cloud init** provisioning), orchestrating every service via **Docker Compose** behind an auto renewing **TLS** reverse proxy, hardened to 3 inbound ports with datastores on a private network

DocMinds | *Python, FastAPI, PostgreSQL + pgvector, Next.js, Docker* | GitHub

- Built a **FastAPI** upload service that reads **19 file extensions** with **PyMuPDF**, **python-docx**, **python-pptx**, **openpyxl** and **BeautifulSoup**, opens ZIP files and reads the documents inside them too, sends scanned PDFs to **Tesseract OCR** when a page has under 100 characters
- Turned each piece of text into a **384 number** vector with the **sentence-transformers** model **all-MiniLM-L6-v2** running on the server, and saved it with its page number in **PostgreSQL** using **pgvector** with an **HNSW** index that keeps search fast as documents grow
- Created the full **RAG** flow so a question becomes a vector the same way, the closest chunks are found by **cosine distance** and dropped if they score too low, and **Llama 3.3 70B** on **Groq** answers only from those chunks and shows the page it used

Online Coding Judge | *Python, FastAPI, PostgreSQL, React, TypeScript, Docker* | GitHub

- Built a full stack coding judge where a user writes **Python, C++ or Java** in a **React** editor and submits it to a **FastAPI** backend that verifies a **JWT** token, loads the problem limits from **PostgreSQL**, and queues the code to run
- Executed the untrusted code inside a fresh **Docker** container with **no network**, a **read only** file system, a non root user, **128 MB** memory, **50% CPU** and a **2s** timeout, then replayed it over sample and hidden tests to return **Accepted, Wrong Answer, Runtime Error or TLE** with time and memory used
- Shipped **37 REST endpoints** across 7 areas covering auth, problems, test cases, submissions, contests and dashboards, so an admin can add a problem with hidden tests while a user registers for a contest, solves it, and sees a **score ranked leaderboard** update live

Technical Skills

Languages: Python, C++, SQL

Backend: FastAPI, REST API Design, WebSockets, Async I/O, Celery Background Workers, Redis Caching, Authentication & Multi-Tenant Access Control

Databases: PostgreSQL, PostGIS, pgvector, Redis, Indexing & Query Optimization (GiST, HNSW)

AI/ML: Google OR-Tools, RAG Pipelines, Sentence-Transformers Embeddings, Vector Search & Semantic Retrieval, LLM Inference (Llama 3.3 on Groq), OCR (Tesseract, PyMuPDF), Computer Vision (OpenCV, dlib), Face Verification & Liveness Detection

DevOps: Docker, Container Sandboxing, AWS EC2, Nginx Reverse Proxy & TLS, GitHub Actions CI, Linux

Frontend: React, Next.js, TypeScript, Tailwind CSS